

A CUBIC LOGARITHMIC UPPER BOUND FOR COMMUTATORS CLOSE TO THE IDENTITY

HYRA

ABSTRACT. Let H be an infinite-dimensional complex Hilbert space and, for $0 < \varepsilon < 1$, define

$$m_H(\varepsilon) := \inf \{ \|D\| \|X\| : D, X \in \mathcal{B}(H), \|[D, X] - \mathbf{1}\| \leq \varepsilon \}.$$

Popa's lower bound is $m_H(\varepsilon) \geq \frac{1}{2} \log(1/\varepsilon)$, while Tao proved the upper bound $m_H(\varepsilon) = O(\log^5(1/\varepsilon))$. We retain Tao's operator-matrix template and replace the worst-case estimate for its linear right inverse by an entrywise supersolution argument adapted to the endpoint forcing in the nonlinear system. The key input is a sharp norm estimate for matrices of the form $\begin{pmatrix} a & b \\ c & a \end{pmatrix}$ and a profile whose square has a controlled discrete concavity defect. This produces solutions of size $O(\delta n)$ with $\delta \asymp n^{-3}$ and yields

$$m_H(\varepsilon) \leq 5.3 \times 10^6 \left(\log \frac{1}{\varepsilon} \right)^3, \quad 0 < \varepsilon \leq \frac{1}{2}.$$

The proof is uniform in ε and applies to arbitrary, not necessarily separable, infinite-dimensional Hilbert spaces.

1. INTRODUCTION

Throughout, H is an infinite-dimensional complex Hilbert space, $\mathcal{B}(H)$ is the algebra of bounded operators on H with its operator norm, $\mathbf{1}$ denotes the identity, and

$$[D, X] := DX - XD.$$

All logarithms without a subscript are natural.

The Wintner–Wielandt theorem states that the identity of a unital Banach algebra cannot be a commutator of two bounded elements [7, 6]. For $\mathcal{B}(H)$, Brown and Percy showed that the identity can nevertheless be approximated arbitrarily well in norm by bounded commutators [2]. The quantitative problem is to determine the least possible growth of the product of the two operator norms. For $0 < \varepsilon < 1$, set

$$(1.1) \quad m_H(\varepsilon) := \inf \{ \|D\| \|X\| : D, X \in \mathcal{B}(H), \|[D, X] - \mathbf{1}\| \leq \varepsilon \}.$$

2020 *Mathematics Subject Classification*. Primary 47A63; Secondary 46L05, 47A20.

Key words and phrases. commutator, bounded operator, Cuntz isometries, nonlinear fixed point, discrete supersolution.

Hyra is an AI research-agent system. The proof originated in Hyra search runs; the accompanying repository contribution records provenance and the exact scope of the Lean formalization.

Popa proved the lower bound

$$(1.2) \quad m_H(\varepsilon) \geq \frac{1}{2} \log \frac{1}{\varepsilon};$$

see [4], and [5] for a short proof. Tao refined the Brown–Pearcy construction and obtained

$$(1.3) \quad m_H(\varepsilon) = O\left(\log^5 \frac{1}{\varepsilon}\right), \quad 0 < \varepsilon \leq \frac{1}{2},$$

see [5]. Krishna and Johnson extended the same exponent to a class of unital C^* -algebras containing $\mathcal{B}(H)$ and the Cuntz algebra $\mathcal{O}_2$ [3]. A publicly available 2026 manuscript, accompanied by a Lean formalization, gives an $O(\log^4(1/\varepsilon))$ refinement [1]. The argument below gives the following cubic logarithmic upper bound.

Theorem 1.1. *Let H be an infinite-dimensional complex Hilbert space. For every $0 < \varepsilon \leq \frac{1}{2}$ there exist $D_\varepsilon, X_\varepsilon \in \mathcal{B}(H)$ such that*

$$(1.4) \quad \|[D_\varepsilon, X_\varepsilon] - \mathbf{1}\| \leq \varepsilon$$

and

$$(1.5) \quad \|D_\varepsilon\| \|X_\varepsilon\| \leq 5.3 \times 10^6 \left(\log \frac{1}{\varepsilon}\right)^3.$$

Consequently,

$$m_H(\varepsilon) = O\left(\log^3 \frac{1}{\varepsilon}\right) \quad (\varepsilon \downarrow 0).$$

The algebraic template is Tao’s. For a matrix size n and a small parameter δ , one constructs $D, X \in M_n(\mathcal{B}(H))$ so that $[D, X] - \mathbf{1}$ is supported in the last column. A nonlinear system is solved to annihilate rows $2, \dots, n$ of that column, leaving only the $(1, n)$ corner. A diagonal similarity then multiplies this corner by $2^{-(n-1)}$, while the norm product is $O(\delta^{-1})$.

The exponent is therefore governed by the largest admissible value of δ . The linear part of the nonlinear system has a right inverse $R = L(I - E)^{-1}$ whose global operator norm is $O(n^2)$. A direct use of this worst-case norm loses two powers of n . The actual forcing, however, is an endpoint spike. We dominate the Neumann series entrywise by a scalar profile ψ satisfying

$$(1.6) \quad s_i + \frac{1}{2} \left\| \begin{pmatrix} \psi_i & \psi_{i-1} \\ \psi_{i+1} & \psi_i \end{pmatrix} \right\| \leq \psi_i.$$

The sharp two-by-two estimate in Lemma 5.2 reduces (1.6) to a discrete concavity inequality for $\Psi_i := \psi_i^2$. The quadratic profile

$$(1.7) \quad \Psi_i = 144\delta^2 i(2n - i)$$

has a constant interior concavity defect and a much larger defect at the right endpoint, exactly where the forcing is concentrated. This gives $\|\beta_i\| = O(\delta n)$, permits $\delta \asymp n^{-3}$, and hence yields the exponent three after taking $n \asymp \log(1/\varepsilon)$.

The proof establishes an upper bound only. In particular, it does not show that the exponent three is optimal; the lower and upper logarithmic exponents remain one and three, respectively.

An accompanying Lean 4 project machine-checks the sharp two-by-two norm estimate, the concavity and supersolution core, the quadratic-profile arithmetic, and the final numerical constants. It does not formalize the operator-valued Neumann-series convergence, the Banach fixed-point construction, or the complete existence theorem in $\mathcal{B}(H)$; those steps remain part of the proof below.

2. OPERATOR-MATRIX PRELIMINARIES

For every finite n , an infinite-dimensional Hilbert space is unitarily isomorphic to its n -fold Hilbert direct sum. Thus there is a unitary $V_n : H^{\oplus n} \rightarrow H$, and conjugation by V_n identifies $M_n(\mathcal{B}(H)) = \mathcal{B}(H^{\oplus n})$ isometrically and $*$ -isomorphically with $\mathcal{B}(H)$. We shall construct the operators in $M_n(\mathcal{B}(H))$ and use this identification at the end.

We record the elementary norm estimates used below.

Lemma 2.1. *Let $M = (M_{ij}) \in M_n(\mathcal{B}(H))$.*

- (1) *If M has only one nonzero entry, at (p, q) , then $\|M\| = \|M_{pq}\|$.*
- (2) *If $M_{ij} = 0$ for $j \neq q$, then*

$$\|M\| \leq \sum_{i=1}^n \|M_{iq}\|.$$

- (3) *If, for a fixed integer k with $1 - n \leq k \leq n - 1$, $M_{ij} = c_i \mathbf{1}_{\{j=i+k\}}$, then*

$$\|M\| = \max_{\substack{1 \leq i \leq n \\ 1 \leq i+k \leq n}} \|c_i\|.$$

- (4) *For $z_{11}, z_{12}, z_{21}, z_{22} \in \mathcal{B}(H)$,*

$$\left\| \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \right\|_{M_2(\mathcal{B}(H))} \leq \left\| \begin{pmatrix} \|z_{11}\| & \|z_{12}\| \\ \|z_{21}\| & \|z_{22}\| \end{pmatrix} \right\|_{M_2(\mathbb{R})}.$$

- (5) *The operator norm of a real 2×2 matrix with nonnegative entries is nondecreasing in each entry.*

Proof. The first statement follows by testing on unit vectors supported in the q -th coordinate and using an approximating sequence for the norm of M_{pq} . The second and third statements follow directly from the action of M on $H^{\oplus n}$; in the third statement one tests on a single valid coordinate and an approximating sequence of unit vectors in H to obtain the reverse inequality.

For the fourth statement, if $h = (h_1, h_2)$, the triangle inequality bounds the norm of each component of Mh by the corresponding component of the scalar matrix of entry norms applied to $(\|h_1\|, \|h_2\|)$. The fifth statement

follows from

$$\|A\|_2 = \sup_{\|x\|_2=\|y\|_2=1} |y^\top Ax|$$

and the observation that, for $A \geq 0$ entrywise, $|y^\top Ax| \leq |y|^\top A|x|$, where $|x|$ and $|y|$ denote coordinatewise absolute values. $\square$

Because $H \cong H \oplus H$, there exist isometries $u, v \in \mathcal{B}(H)$ with orthogonal ranges whose range projections sum to the identity:

$$(2.1) \quad u^*u = v^*v = \mathbf{1}, \quad u^*v = v^*u = 0, \quad uu^* + vv^* = \mathbf{1}.$$

For example, choose a unitary $W : H \oplus H \rightarrow H$ and put $u\xi = W(\xi, 0)$ and $v\xi = W(0, \xi)$. This construction works equally in the nonseparable case.

Lemma 2.2. *The map $U : H \oplus H \rightarrow H$ given by $U(\xi, \eta) = u\xi + v\eta$ is unitary. Consequently,*

$$(2.2) \quad \begin{aligned} & \|uz_{11}u^* + uz_{12}v^* + vz_{21}u^* + vz_{22}v^*\| \\ &= \left\| \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} \right\|_{M_2(\mathcal{B}(H))}. \end{aligned}$$

In particular,

$$(2.3) \quad \|xv^* + yu^*\| \leq (\|x\|^2 + \|y\|^2)^{1/2}.$$

Proof. The relations (2.1) show that U is an onto isometry and that the operator on the left of (2.2) is $U \begin{pmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \end{pmatrix} U^*$. For (2.3), use

$$(xv^* + yu^*)(xv^* + yu^*)^* = xx^* + yy^*.$$

$\square$

3. THE COMMUTATOR TEMPLATE AND DIAGONAL SIMILARITY

Fix an integer $n \geq 3$, a number $\delta > 0$, and operators $\beta_1, \dots, \beta_n \in \mathcal{B}(H)$. Set $\beta_0 = \beta_{n+1} = 0$. Define $X, D \in M_n(\mathcal{B}(H))$ by

$$(3.1) \quad X_{ij} = \mathbf{1}\mathbf{1}_{\{i=j+1\}} + \beta_i\mathbf{1}_{\{j=n\}},$$

$$(3.2) \quad D_{ij} = \delta^{-1}u\mathbf{1}_{\{i=j+1\}} + \delta^{-1}v\mathbf{1}_{\{i=j\}} + i\mathbf{1}\mathbf{1}_{\{j=i+1\}} + \delta^{-1}\beta_i u\mathbf{1}_{\{j=n\}}.$$

When two conditions in (3.2) hold simultaneously, the corresponding terms are added.

Lemma 3.1 (Exact commutator identity). *For $1 \leq i, j \leq n$,*

$$(3.3) \quad \begin{aligned} [D, X]_{ij} &= \mathbf{1}\mathbf{1}_{\{i=j\}} \\ &+ [\delta^{-1}([v, \beta_i] + [u, \beta_{i-1}] + \beta_i[u, \beta_n]) + i\beta_{i+1} - n\mathbf{1}\mathbf{1}_{\{i=n\}}] \mathbf{1}_{\{j=n\}}. \end{aligned}$$

Thus $[D, X] - \mathbf{1}$ is supported in the last column.

Proof. Let $S_{ij} = \mathbf{1}\mathbf{1}_{\{i=j+1\}}$, let $B_{ij} = \beta_i \mathbf{1}_{\{j=n\}}$, and let $N_{ij} = i \mathbf{1}\mathbf{1}_{\{j=i+1\}}$. Then $X = S + B$ and

$$(3.4) \quad D = \delta^{-1} \text{diag}(v) + \delta^{-1} X \text{diag}(u) + N.$$

The commutator $[\text{diag}(v), X]$ vanishes off the last column, where its row- i entry is $[v, \beta_i]$. Moreover,

$$[X \text{diag}(u), X] = X[\text{diag}(u), X],$$

which also vanishes off the last column, where its row- i entry is $[u, \beta_{i-1}] + \beta_i[u, \beta_n]$.

Finally, direct multiplication gives

$$[N, X]_{ij} = \mathbf{1}\mathbf{1}_{\{i=j\}} + (i\beta_{i+1} - n\mathbf{1}_{\{i=n\}})\mathbf{1}_{\{j=n\}}.$$

Adding the three contributions in (3.4) proves (3.3). $\square$

We say that $\beta = (\beta_1, \dots, \beta_n)$ solves the nonlinear system if

$$(3.5) \quad [v, \beta_i] + [u, \beta_{i-1}] + \beta_i[u, \beta_n] + \delta i \beta_{i+1} = \delta n \mathbf{1}_{\{i=n\}}, \quad 2 \leq i \leq n.$$

If (3.5) holds, then $[D, X] - \mathbf{1}$ has only one possibly nonzero entry, namely

$$(3.6) \quad Z_{1n} = \delta^{-1}([v, \beta_1] + \beta_1[u, \beta_n]) + \beta_2.$$

For $0 < \mu \leq 1$, put

$$(3.7) \quad \begin{aligned} S_\mu &= \text{diag}(\mu^{n-1}\mathbf{1}, \mu^{n-2}\mathbf{1}, \dots, \mathbf{1}), \\ D_\mu &= \mu^{-1} S_\mu D S_\mu^{-1}, \quad X_\mu = \mu S_\mu X S_\mu^{-1}. \end{aligned}$$

Proposition 3.2 (Diagonal similarity). *For arbitrary $\beta_1, \dots, \beta_n$,*

$$(3.8) \quad [D_\mu, X_\mu] = S_\mu [D, X] S_\mu^{-1},$$

$$(3.9) \quad \|X_\mu\| \leq 1 + \sum_{i=1}^n \mu^{n-i+1} \|\beta_i\|,$$

and

$$(3.10) \quad \|D_\mu\| \leq \frac{1}{\mu\delta} + \frac{1}{\mu^2\delta} + (n-1) + \frac{1}{\delta} \sum_{i=1}^n \mu^{n-i-1} \|\beta_i\|.$$

If β solves (3.5), then

$$(3.11) \quad \|[D_\mu, X_\mu] - \mathbf{1}\| = \mu^{n-1} \|Z_{1n}\|.$$

Proof. The scalar factors in (3.7) cancel in the commutator, which gives (3.8). Conjugation by S_μ multiplies an (i, j) entry by μ^{j-i} . Hence X_μ is the unweighted subdiagonal shift plus a last column with row- i entry $\mu^{n-i+1}\beta_i$, proving (3.9) by Lemma 2.1.

Similarly, the four parts of D_μ are a diagonal of norm $(\mu\delta)^{-1}$, a subdiagonal of norm $(\mu^2\delta)^{-1}$, a weighted superdiagonal of norm $n-1$, and a last column with row- i entry $\delta^{-1}\mu^{n-i-1}\beta_i u$. This gives (3.10). If (3.5) holds, then (3.6) and the first part of Lemma 2.1 give (3.11). $\square$

4. A RIGHT INVERSE FOR THE LINEARIZED SYSTEM

Let

$$Y := \mathcal{B}(H)^{n-1}, \quad Z := \mathcal{B}(H)^n,$$

where elements of Y are indexed by $2, \dots, n$, and both spaces carry the maximum norm. For $x \in Y$ we use the boundary conventions $x_1 = x_{n+1} = 0$. Define $T : Z \rightarrow Y$, $L : Y \rightarrow Z$, and $E : Y \rightarrow Y$ by

$$(4.1) \quad (Tb)_i = [v, b_i] + [u, b_{i-1}], \quad 2 \leq i \leq n,$$

$$(4.2) \quad (Lx)_i = -\frac{1}{2}(x_i v^* + x_{i+1} u^*), \quad 1 \leq i \leq n,$$

$$(4.3) \quad (Ex)_i = \frac{1}{2}(vx_i v^* + vx_{i+1} u^* + ux_{i-1} v^* + ux_i u^*), \quad 2 \leq i \leq n.$$

Lemma 4.1. *One has*

$$(4.4) \quad TL = I_Y - E.$$

Proof. For $2 \leq i \leq n$, the Cuntz relations give

$$\begin{aligned} [v, (Lx)_i] &= -\frac{1}{2}(vx_i v^* + vx_{i+1} u^*) + \frac{1}{2}x_i, \\ [u, (Lx)_{i-1}] &= -\frac{1}{2}(ux_{i-1} v^* + ux_i u^*) + \frac{1}{2}x_i. \end{aligned}$$

Their sum is $x_i - (Ex)_i$. □

Lemma 4.2 (Weighted contraction). *For $x \in Y$, define*

$$(4.5) \quad \|x\|' := \max_{2 \leq i \leq n} \left(2 - \frac{i^2}{n^2}\right)^{-1/2} \|x_i\|.$$

Then

$$(4.6) \quad \frac{1}{\sqrt{2}} \|x\|_Y \leq \|x\|' \leq \|x\|_Y$$

and

$$(4.7) \quad \|Ex\|' \leq \left(1 - \frac{1}{8n^2}\right) \|x\|'.$$

Proof. The norm equivalence is immediate from $1 \leq 2 - i^2/n^2 \leq 2$ for $2 \leq i \leq n$. By homogeneity, assume $\|x\|' \leq 1$, so $\|x_i\|^2 \leq 2 - i^2/n^2$. This inequality also holds for $i = 1, n+1$ because $x_1 = x_{n+1} = 0$; note that $2 - (n+1)^2/n^2 > 0$ for $n \geq 3$.

By Lemma 2.2,

$$(4.8) \quad \|(Ex)_i\| = \frac{1}{2} \left\| \begin{pmatrix} x_i & x_{i-1} \\ x_{i+1} & x_i \end{pmatrix} \right\|_{M_2(\mathcal{B}(H))}.$$

Using Lemma 2.1 and then the Frobenius norm of the scalar matrix of entry norms,

$$\|(Ex)_i\|^2 \leq \frac{1}{4} (2\|x_i\|^2 + \|x_{i-1}\|^2 + \|x_{i+1}\|^2)$$

$$\begin{aligned}
&\leq 2 - \frac{i^2}{n^2} - \frac{1}{2n^2} \\
&\leq \left(1 - \frac{1}{4n^2}\right) \left(2 - \frac{i^2}{n^2}\right) \\
&\leq \left(1 - \frac{1}{8n^2}\right)^2 \left(2 - \frac{i^2}{n^2}\right).
\end{aligned}$$

Taking square roots and maximizing over i proves (4.7). $\square$

Proposition 4.3 (Right inverse). *The operator $I_Y - E$ is invertible, and*

$$(4.9) \quad R := L(I_Y - E)^{-1} : Y \rightarrow Z$$

is a right inverse for T . Moreover,

$$(4.10) \quad \|R\|_{Y \rightarrow Z} \leq 8\sqrt{2}n^2.$$

Proof. The Neumann series converges in the primed norm by Lemma 4.2, and

$$\|(I_Y - E)^{-1}\|' \leq 8n^2.$$

Thus Lemma 4.1 gives $TR = I_Y$. If $\|x\|_Y \leq 1$ and $y = (I_Y - E)^{-1}x$, then $\|y_i\| \leq 8\sqrt{2}n^2$. By (4.2) and (2.3),

$$\|(Rx)_i\| \leq \frac{1}{2}(\|y_i\|^2 + \|y_{i+1}\|^2)^{1/2} \leq 8n^2,$$

which is stronger than (4.10). $\square$

Remark 4.4. The n^2 scale in (4.10) agrees with the spectral-gap scale of the corresponding scalar Dirichlet averaging operator. We do not need a matching lower bound for R ; the improvement below comes from estimating the specific structured data fed into R , rather than its global norm.

5. A MONOTONE SUPERSOLUTION PRINCIPLE

For a nonnegative vector $p = (p_1, \dots, p_{n+1})$, define

$$(5.1) \quad (Ap)_i := \frac{1}{2} \left\| \begin{pmatrix} p_i & p_{i-1} \\ p_{i+1} & p_i \end{pmatrix} \right\|_{M_2(\mathbb{R})}, \quad 2 \leq i \leq n.$$

By the last part of Lemma 2.1, A is monotone in every coordinate.

Lemma 5.1 (Majorant property). *Let $x \in Y$ and put $p_j = \|x_j\|$, with $p_1 = p_{n+1} = 0$. Then*

$$\|(Ex)_i\| \leq (Ap)_i, \quad 2 \leq i \leq n.$$

Proof. Combine (4.8) with the fourth part of Lemma 2.1. $\square$

The following elementary inequality is the key quantitative input.

Lemma 5.2 (Sharp two-by-two estimate). *Let $a, b, c \geq 0$ and assume $b^2 + c^2 \leq 2a^2$. Then*

$$(5.2) \quad \left\| \begin{pmatrix} a & b \\ c & a \end{pmatrix} \right\|_{M_2(\mathbb{R})} \leq a + \sqrt{\frac{b^2 + c^2}{2}}.$$

Consequently, if $a > 0$ and $d := a^2 - (b^2 + c^2)/2 \geq 0$, then

$$(5.3) \quad \frac{1}{2} \left\| \begin{pmatrix} a & b \\ c & a \end{pmatrix} \right\| \leq a - \frac{d}{4a}.$$

Equality in (5.2) is possible, for example when $b = c$.

Proof. Put $P = a^2$, $Q = b^2$, and $R = c^2$. A direct computation of the largest eigenvalue of $M^\top M$, where $M = \begin{pmatrix} a & b \\ c & a \end{pmatrix}$, gives

$$(5.4) \quad \|M\|^2 = P + \frac{Q+R}{2} + \sqrt{\frac{(Q-R)^2}{4} + P(b+c)^2}.$$

Let $W = (b+c)^2$ and $e = (b-c)^2$. Then $W+e = 2(Q+R)$ and $(Q-R)^2 = eW$. Since $Q+R \leq 2P$, one has $W \leq 4P$, and therefore

$$\frac{eW}{4} + PW \leq eP + PW = P(e+W) = 2P(Q+R).$$

Substitution into (5.4) gives

$$\|M\|^2 \leq P + \frac{Q+R}{2} + \sqrt{2P(Q+R)} = \left(\sqrt{P} + \sqrt{\frac{Q+R}{2}} \right)^2,$$

which proves (5.2).

For (5.3), write $(Q+R)/2 = P - d$ and use the tangent-line bound $\sqrt{P-d} \leq \sqrt{P} - d/(2\sqrt{P})$. $\square$

Proposition 5.3 (Supersolution bound). *Let $s = (s_i)_{i=2}^n$ be nonnegative and let $\psi = (\psi_1, \dots, \psi_{n+1})$ be nonnegative. Suppose*

$$(5.5) \quad s_i + (A\psi)_i \leq \psi_i, \quad 2 \leq i \leq n.$$

If $x \in Y$ satisfies $\|x_i\| \leq s_i$, then

$$(5.6) \quad \left\| \left(\sum_{k=0}^K E^k x \right)_i \right\| \leq \psi_i \quad (K \geq 0, 2 \leq i \leq n),$$

$$(5.7) \quad \|((I_Y - E)^{-1}x)_i\| \leq \psi_i,$$

and

$$(5.8) \quad \|(Rx)_i\| \leq \frac{1}{2}(\psi_i^2 + \psi_{i+1}^2)^{1/2}, \quad 1 \leq i \leq n.$$

Proof. Set $y^{[K]} = \sum_{k=0}^K E^k x$. The case $K = 0$ follows from $s_i \leq \psi_i$. If (5.6) holds for K , then $y^{[K+1]} = x + Ey^{[K]}$, and the majorant property together with the monotonicity of A gives

$$\left\| y_i^{[K+1]} \right\| \leq s_i + (A\psi)_i \leq \psi_i.$$

This proves (5.6) by induction. Passing to the Neumann series limit proves (5.7). Finally, apply (2.3) to $R = L(I_Y - E)^{-1}$. $\square$

Corollary 5.4 (Concavity criterion). *Assume $\psi_i > 0$ for $1 \leq i \leq n$, put $\psi_{n+1} = 0$, and define $\Psi_i = \psi_i^2$ and*

$$(5.9) \quad d_i := \Psi_i - \frac{\Psi_{i-1} + \Psi_{i+1}}{2}, \quad 2 \leq i \leq n.$$

If

$$(5.10) \quad d_i \geq 0 \quad \text{and} \quad 4\psi_i s_i \leq d_i, \quad 2 \leq i \leq n,$$

then (5.5) holds, and all conclusions of Proposition 5.3 follow.

Proof. Apply (5.3) with $a = \psi_i$, $b = \psi_{i-1}$, and $c = \psi_{i+1}$. The first condition in (5.10) is exactly $b^2 + c^2 \leq 2a^2$, and it yields

$$(A\psi)_i \leq \psi_i - \frac{d_i}{4\psi_i} \leq \psi_i - s_i.$$

$\square$

6. SOLVING THE NONLINEAR SYSTEM

From now on let

$$(6.1) \quad n \geq 9, \quad \delta := \frac{1}{400n^3}.$$

Define

$$(6.2) \quad \Psi_i := 144\delta^2 i(2n - i), \quad 1 \leq i \leq n, \quad \Psi_{n+1} := 0,$$

$$(6.3) \quad \psi_i := \sqrt{\Psi_i} = 12\delta\sqrt{i(2n - i)}, \quad 1 \leq i \leq n, \quad \psi_{n+1} := 0,$$

$$(6.4) \quad \bar{b}_i := \frac{1}{2}\sqrt{\Psi_i + \Psi_{i+1}}, \quad 1 \leq i \leq n, \quad \bar{b}_{n+1} := 0,$$

$$(6.5) \quad s_i := \delta n \mathbf{1}_{\{i=n\}} + 2\bar{b}_i \bar{b}_n + \delta i \bar{b}_{i+1}, \quad 2 \leq i \leq n.$$

Lemma 6.1 (Profile estimates). *The following estimates hold:*

$$(6.6) \quad \psi_i \leq 12\delta n,$$

$$(6.7) \quad \psi_i \leq 12\sqrt{2}\delta\sqrt{ni} < 17\delta\sqrt{ni},$$

$$(6.8) \quad \bar{b}_i \leq 6\sqrt{2}\delta n,$$

$$(6.9) \quad \bar{b}_i \leq 15\delta\sqrt{ni}, \quad \bar{b}_n = 6\delta n,$$

$$(6.10) \quad d_i = 144\delta^2, \quad 2 \leq i \leq n - 1,$$

$$(6.11) \quad d_n = 72\delta^2(n^2 + 1),$$

$$(6.12) \quad \bar{b}_1 \leq 6\sqrt{6}\delta\sqrt{n}, \quad \bar{b}_2 \leq 6\sqrt{10}\delta\sqrt{n}.$$

Proof. The function $i(2n - i) = n^2 - (n - i)^2$ is increasing on $[0, n]$ and bounded by n^2 , giving (6.6). Since $i(2n - i) \leq 2ni$, one obtains (6.7).

From $4\bar{b}_i^2 = \Psi_i + \Psi_{i+1} \leq 288\delta^2 n^2$ one obtains (6.8). For $i < n$,

$$4\bar{b}_i^2 \leq 144\delta^2(2ni + 2n(i + 1)) \leq 864\delta^2 ni,$$

which implies $\bar{b}_i \leq \sqrt{216}\delta\sqrt{ni} < 15\delta\sqrt{ni}$. The formula for $\bar{b}_n$ follows from $\Psi_{n+1} = 0$.

For $2 \leq i < n$, the second difference of $i(2n - i)$ is -2 , hence $d_i = 144\delta^2$. At $i = n$,

$$d_n = 144\delta^2 n^2 - \frac{1}{2} 144\delta^2(n^2 - 1) = 72\delta^2(n^2 + 1).$$

Finally,

$$\Psi_1 + \Psi_2 = 144\delta^2(6n - 5) \leq 864\delta^2 n$$

and

$$\Psi_2 + \Psi_3 = 144\delta^2(10n - 13) \leq 1440\delta^2 n,$$

which give (6.12). $\square$

Lemma 6.2 (Verification of the supersolution condition). *For every $2 \leq i \leq n$,*

$$(6.13) \quad 4\psi_i s_i \leq d_i.$$

Consequently, the pair (s, ψ) satisfies (5.5).

Proof. For $i = n$, (6.5) and (6.9) give

$$s_n = \delta n + 72\delta^2 n^2, \quad \psi_n = 12\delta n.$$

Therefore

$$4\psi_n s_n = 48\delta^2 n^2 + 3456\delta^3 n^3 = \left(48n^2 + \frac{3456}{400}\right)\delta^2 \leq 72(n^2 + 1)\delta^2 = d_n.$$

Now let $2 \leq i \leq n - 1$. By (6.9),

$$\begin{aligned} s_i &\leq 2(15\delta\sqrt{ni})(6\delta n) + \delta i(15\sqrt{2}\delta\sqrt{ni}) \\ &\leq 201.3\delta^2 n\sqrt{ni}. \end{aligned}$$

Together with (6.7), this gives

$$4\psi_i s_i \leq 13689\delta^3 n^2 i \leq 13689\delta^3 n^3 = \frac{13689}{400}\delta^2 < 144\delta^2 = d_i.$$

The conclusion follows from Corollary 5.4. $\square$

Proposition 6.3 (Nonlinear solution). *For every $n \geq 9$ and $\delta = 1/(400n^3)$, the system (3.5) has a solution $\beta = (\beta_1, \dots, \beta_n) \in \mathcal{B}(H)^n$ such that*

$$(6.14) \quad \|\beta_i\| \leq \bar{b}_i, \quad 1 \leq i \leq n.$$

In particular,

$$(6.15) \quad \begin{aligned} \max_i \|\beta_i\| &\leq 6\sqrt{2}\delta n, & \|\beta_n\| &\leq 6\delta n, \\ \|\beta_1\| &\leq 6\sqrt{6}\delta\sqrt{n}, & \|\beta_2\| &\leq 6\sqrt{10}\delta\sqrt{n}. \end{aligned}$$

Proof. Let

$$Q := \{y \in Y : \|y_i\| \leq s_i \text{ for } 2 \leq i \leq n\}.$$

This is a nonempty closed subset of the Banach space Y , hence complete. Define $\Phi : Y \rightarrow Y$ by

$$(6.16) \quad \Phi(y)_i := \delta n \mathbf{1}_{\{i=n\}} - (Ry)_i[u, (Ry)_n] - \delta i (Ry)_{i+1}, \quad 2 \leq i \leq n,$$

where $(Ry)_{n+1} = 0$.

If $y \in Q$ and $b = Ry$, then Lemma 6.2 and Proposition 5.3 give

$$\|b_i\| \leq \frac{1}{2} \sqrt{\psi_i^2 + \psi_{i+1}^2} = \bar{b}_i.$$

Hence

$$\|\Phi(y)_i\| \leq \delta n \mathbf{1}_{\{i=n\}} + 2\bar{b}_i \bar{b}_n + \delta i \bar{b}_{i+1} = s_i,$$

so $\Phi(Q) \subseteq Q$.

For $y, y' \in Q$, put $b = Ry$ and $b' = Ry'$. Using the preceding profile estimate together with (6.8) and $\bar{b}_n = 6\delta n$, one obtains

$$\begin{aligned} \|\Phi(y) - \Phi(y')\|_Y &\leq (12\delta n + 12\sqrt{2}\delta n + \delta n) \|b - b'\|_Z \\ &\leq (13 + 12\sqrt{2})\delta n 8\sqrt{2}n^2 \|y - y'\|_Y \\ &= \frac{192 + 104\sqrt{2}}{400} \|y - y'\|_Y < 0.85 \|y - y'\|_Y. \end{aligned}$$

Thus Φ is a contraction of Q . Let $y^* \in Q$ be its unique fixed point and put $\beta = Ry^*$. Since $TR = I_Y$,

$$T\beta = y^* = \Phi(y^*),$$

which is exactly (3.5). The bounds follow from the profile estimate above and Lemma 6.1. $\square$

7. NORM ESTIMATES AND COMPLETION OF THE PROOF

Take $\mu = 1/2$ in Proposition 3.2, and let D_μ, X_μ be built from the solution in Proposition 6.3.

Proposition 7.1. *For $n \geq 9$ and $\delta = 1/(400n^3)$,*

$$(7.1) \quad \|X_\mu\| \leq 1.00027,$$

$$(7.2) \quad \|D_\mu\| \leq 2401n^3,$$

$$(7.3) \quad \|D_\mu\| \|X_\mu\| \leq 2402n^3,$$

$$(7.4) \quad \|[D_\mu, X_\mu] - \mathbf{1}\| \leq 59\sqrt{n} 2^{-n}.$$

Proof. Let $\bar{b} = \max_i \bar{b}_i$. By (6.8), $\bar{b} \leq 6\sqrt{2}\delta n$. Since $\sum_{i=1}^n 2^{-(n-i+1)} < 1$, (3.9) gives

$$\|X_\mu\| \leq 1 + \bar{b} \leq 1 + \frac{6\sqrt{2}}{400n^2} \leq 1.00027.$$

For D_μ , (3.10) and $\sum_{i=1}^n 2^{i+1-n} < 4$ yield

$$\begin{aligned} \|D_\mu\| &\leq \frac{6}{\delta} + (n-1) + \frac{4\bar{b}}{\delta} \\ &\leq 2400n^3 + (1 + 24\sqrt{2})n \\ &< 2400n^3 + 35n \leq 2401n^3. \end{aligned}$$

The product estimate follows because $2401 \cdot 1.00027 < 2402$.

It remains to estimate the surviving corner in (3.6). By (6.15),

$$\begin{aligned} \|Z_{1n}\| &\leq \frac{2}{\delta} \|\beta_1\| + \frac{2}{\delta} \|\beta_1\| \|\beta_n\| + \|\beta_2\| \\ &\leq \sqrt{n} \left(12\sqrt{6} + 72\sqrt{6}\delta n + 6\sqrt{10}\delta \right). \end{aligned}$$

For $n \geq 9$, one has $\delta n \leq 1/32400$ and $\delta \leq 1/291600$. Using $\sqrt{6} < 2.4495$ and $\sqrt{10} < 3.1623$, the coefficient in parentheses is strictly less than 29.4. Hence, by (3.11),

$$\|[D_\mu, X_\mu] - \mathbf{1}\| \leq 29.4\sqrt{n} 2^{-(n-1)} < 59\sqrt{n} 2^{-n}.$$

□

Proof of Theorem 1.1. Let

$$t := \log_2 \frac{1}{\varepsilon} \geq 1, \quad n := \max \{9, \lceil 2t \rceil + 6\}.$$

Construct $D_\mu, X_\mu \in M_n(\mathcal{B}(H))$ as above and transport them to $\mathcal{B}(H)$ through any unitary $H^{\oplus n} \rightarrow H$. This preserves norms, products, the identity, and commutators.

We first prove the error bound. If $n = 9$, then $\lceil 2t \rceil \leq 3$, so $t \leq 3/2$, and

$$59\sqrt{n} 2^{-n} = \frac{177}{512} < 2^{-3/2} \leq 2^{-t} = \varepsilon.$$

If $n \geq 10$, then $n = \lceil 2t \rceil + 6$, so $t \leq (n-6)/2$. Therefore

$$\frac{59\sqrt{n} 2^{-n}}{\varepsilon} = 59\sqrt{n} 2^{t-n} \leq \frac{59\sqrt{n}}{8 \cdot 2^{n/2}} \leq \frac{59\sqrt{10}}{256} < 1,$$

where the middle expression decreases for integers $n \geq 10$. Thus (1.4) follows from (7.4).

For the norm product, note that $n \leq 9t$: this is immediate when $n = 9$, and otherwise $n \leq 2t + 7 \leq 9t$ because $t \geq 1$. Writing $L = \log(1/\varepsilon) = t \log 2$, (7.3) gives

$$\begin{aligned} \|D_\varepsilon\| \|X_\varepsilon\| &\leq 2402n^3 \\ &\leq 2402 \left(\frac{9}{\log 2} \right)^3 L^3 \end{aligned}$$

$$< 5.3 \times 10^6 L^3.$$

This proves (1.5) and completes the proof. $\square$

8. FURTHER REMARKS

Remark 8.1 (Finite-dimensional obstruction). If $\dim H = N < \infty$, then $\operatorname{tr}[D, X] = 0$, so

$$\|[D, X] - \mathbf{1}\| \geq \frac{1}{N} |\operatorname{tr}([D, X] - \mathbf{1})| = 1.$$

Thus the approximation problem is genuinely infinite-dimensional. The proof uses infinite dimensionality precisely through the identifications $H \cong H^{\oplus n}$ and the Cuntz relations (2.1).

Remark 8.2 (Effectivity). The solution in Proposition 6.3 is obtained by a Neumann series and a Picard iteration with an explicit contraction factor below 0.85. Truncating both procedures therefore gives finite noncommutative polynomials in u, v, u^*, v^* with explicitly controlled residuals. In particular, the construction is algorithmic; the limiting formulation is used only to keep the main proof concise.

Remark 8.3 (Source of the exponent). The profile estimate gives $\|\beta_i\| = O(\delta n)$. The quadratic and shift terms in (3.5) are both of size $O(\delta^2 n^2)$. A profile of height $O(\delta n)$ over a chain of length n has an interior concavity budget of order δ^2 , so absorption requires $\delta^3 n^3 \lesssim \delta^2$, equivalently $\delta \lesssim n^{-3}$. Since $\|D\| \|X\| = O(\delta^{-1})$ and $n \asymp \log(1/\varepsilon)$, this produces the cubic logarithm.

Remark 8.4 (What remains open). Combining (1.2) with Theorem 1.1 leaves

$$\frac{1}{2} \log \frac{1}{\varepsilon} \leq m_H(\varepsilon) \leq C \left(\log \frac{1}{\varepsilon} \right)^3.$$

No lower-bound improvement is claimed here. Reaching the conjectural linear logarithmic scale would require a further structural gain beyond the present endpoint supersolution estimate.

REFERENCES

- [1] B. Bilich, *An $O(\log^4(1/\varepsilon))$ refinement of Tao's construction of commutators close to the identity*, unpublished manuscript and Lean formalization, 2026, <https://github.com/bilichboris/TaoCommutators>.
- [2] A. Brown and C. Pearcy, *Structure of commutators of operators*, Ann. of Math. (2) **82** (1965), 112–127.
- [3] K. M. Krishna and P. S. Johnson, *Commutators close to the identity in unital C^* -algebras*, Proc. Indian Acad. Sci. Math. Sci. **132** (2022), Paper No. 11.
- [4] S. Popa, *On commutators in properly infinite W^* -algebras*, in *Invariant Subspaces and Other Topics*, Operator Theory: Advances and Applications, vol. 6, Birkhäuser, Basel, 1982, pp. 195–207.
- [5] T. Tao, *Commutators close to the identity*, J. Operator Theory **82** (2019), no. 2, 369–382.
- [6] H. Wielandt, *Über die Unbeschränktheit der Operatoren der Quantenmechanik*, Math. Ann. **121** (1949/50), 21.

- [7] A. Wintner, *The unboundedness of quantum-mechanical matrices*, Phys. Rev. (2) **71** (1947), 738–739.